BIMM-143: INTRODUCTION TO BIOINFORMATICS

The find-a-gene project assignment
http://thegrantlab.org/bimm143
Dr. Barry Grant

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Overview:

The find-a-gene project is a required assignment for BIMM-143. You should prepare a written report in **PDF** format that has responses to each question labeled **[Q1] - [Q10]** below. You may wish to consult the scoring rubric at the end of this document and the example report provided online (note that the example report is from a *previous quarter* and the questions may differ).

The objective with this assignment is for you to demonstrate your grasp of database searching, sequence analysis, structure analysis and the R environment that we have covered in class.

Due Date:

Your responses to questions Q1-Q4 are due at 12pm on the **Monday of Week 5** (see the Assignments and Grading section of our website for details). Note that these first set of answers can be obtained very quickly (at best within 15 or 20 minutes), so if you don't succeed at first, just keep trying.

The complete assignment, including responses to all questions, is due at 12pm on the **Monday of Week 10**.

Submission instructions:

Your report formatted as a **PDF document** should be uploaded to ***GradeScope***. Please make sure to include your UCSD email and PID number on the first page.

Be sure to include your UCSD email and PID number on the first page of your report.

Submit your preliminary report with answers to Q1-Q4 as soon as you can so we can determine

if you have found a novel gene. Submit this preliminary report as one document with screen shots of the results inserted appropriately.

See the demonstration report linked to on the course website for an example of format. Note again that example questions may differ. I will indicate on GradeScope my decision (1pt indicating all is good, 0pts revisions required). You should proceed with subsequent questions only after we are sure you have found a novel gene (and thus be successful in the later stages of the project).

For the final report add your results for Q5-Q10 to the preliminary report and submit the final document containing your results for all questions.
Please do not send only Q5-Q10 answers as the final report.
Questions:

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as its function is known.

If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Protein name: Ubiquitin carboxyl-terminal hydrolase 36 isoform 2

Species: Homo sapiens

Accession number: NP_001372103.1

Function: Removes ubiquitin from target proteins, involved in ribosomal RNA processing, protein stabilization, autophagy regulation, cancer progression.

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Also include the output of that BLAST search in your document. If appropriate, change the font to `Courier size 10` so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press ⌘-shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called `Screen Shot [].png` in your Desktop directory). It is **not** necessary to print out all of the blast results if there are many pages.

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to

be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result.

If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

Method: TBLASTN search against oak trees

Database: est

Organism: English oak (taxid:38942)

Chosen match: Accession FP036344, 630 bp Quercus robur cDNA clone

Your search is limited to records **include: English oak (taxid:38942) ; and exclude: models (XM/XP)**

Job Title	NP_001372103:ubiquitin carboxyl-terminal hydrolase...
RID	S1VTJX13014 *Search expires on 02-04 01:38 am* Download All
Program	TBLASTN Citation
Database	est See details
Query ID	NP_001372103.1
Description	ubiquitin carboxyl-terminal hydrolase 36 isoform 2 [Homo ...
Molecule type	amino acid
Query Length	1123
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *24 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
FP036344 LG0AAB Quercus robur cDNA clone LG0AAB24YJ22RM1 similar to UBP15_ARATH Ubiquitin carboxyl-ter...	Quercus robur	132	132	17%	1e-34	37.13%	630	FP036344.1
FP068554 LG0AAD Quercus robur cDNA clone LG0AAD24YG19RM1 similar to UBP17_ARATH Ubiquitin carboxyl-ter...	Quercus robur	100	165	21%	2e-23	30.10%	725	FP068554.1
FR626065 WZ0AQRAS Quercus robur cDNA clone WZ0AQRAS15YL05CM1, mRNA sequence	Quercus robur	97.4	97.4	11%	1e-22	37.80%	590	FR626065.1
FP027236 LG0AAA Quercus robur cDNA clone LG0AAA2YM24RM1 similar to UBP25_ARATH Ubiquitin carboxyl-term...	Quercus robur	97.1	97.1	9%	3e-22	43.56%	677	FP027236.1

Download GenBank Graphics

FP036344 LG0AAB Quercus robur cDNA clone LG0AAB24YJ22RM1 similar to UBP15_ARATH Ubiqu
Arabidopsis thaliana, mRNA sequence

Sequence ID: FP036344.1 Length: 630 Number of Matches: 1

Range 1: 38 to 628 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
132 bits(331)	1e-34	Composition-based stats.	82/202(41%)	113/202(55%)	13/202(6%)	+2

```
Query  123  GLHNLGNTCFLNATIQCLTYTPPLANYLLSKEHARSCHQGSFCMLCVMQNHIVQAFANSG  182
            GL N GN+C+ NA +QCLT T PL  YL+ + H+R+C    +C++C ++ H+V    + G
Sbjct  38   GLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKGWCLMCELEQHVVMLRESGG  217

Query  183  NAIKPVSFIRDLKKIARHFRF----GNQEDAHEFLRYTIDAMQKACLNGCA---KLDRQT  235
                P+S  R L  I R   F    G+QEDAH+FLR  I +MQ  CL G     K+D
Sbjct  218  ----PLSLSRILMHI-RSINFQIGDGSQEDAHDFLRLIIASMQSICLEGLGGEKKVDPIL  382

Query  236  QATTLVHQIFGGYLrsrvkcsvcksvsDTYDPYLDVALEIRQ-AANIVRALELFVKADVL  294
            Q TT +  IFGG LRS+VKC +C   S+ Y+  +D+ LEI     ++  AL  F   + L
Sbjct  383  QETTFIQHIFGGRLRSKVKCMICHQESERYENMMDLTLEIYGWVESLEDALTQFTATEDL  562

Query  295  SGENAYMCAKCKKKVPASKRFT  316
             GEN Y C +C   V A K  +
Sbjct  563  DGENMYRCGRCATYVRAKKELS  628
```

[Q3] Gather information about this "novel" **protein**. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format. Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as *S. cerevisiae*, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

>Quercus robur sequence
FFQCEVFDLSPRGLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKG
WCLMCELEQHVVMLRESGGPLSLSRILMHIRSINFQIGDGSQEDAHDFLRLIIA
SMQSICLEGLGGEKKVDPILQETTFIQHIFGGRLRSKVKCMICHQESERYENM
MDLTLEIYGWVESLEDALTQFTATEDLDGENMYRCGRCATYVRAKKELS

Species: Quercus robur (English oak)
Name: Quercus robur cDNA clone LG0AAB24YJ22RM1 similar to UBP15_ARATH Ubiquitin carboxyl-terminal hydrolase 15

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

- If there is a match with 100% amino acid identity to a protein in the database, from the

same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.

- If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.
- If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.
- If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

BLASTP search against ClusteredNR database

Top match: Castanea Dentata (American chestnut)

Standard Protein BLAST

blastn | **blastp** | blastx | tblastn | tblastx

BLASTP programs search protein databases using a protei

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) Clear

```
>Quercus robur protein
FFQCEVFDLSPRGLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKGW
CLMCELEQHVVMLRES
GGPLSLSRILMHIRSINFQIGDGSQEDAHDFLRLIIASMQSICLEGLGGEKKVDPI
```

Query subrange

From

To

Or, upload file Choose File No file chosen

Job Title Quercus robur protein

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database ClusteredNR (nr_cluster_seq)

Organism Optional

Enter organism name or id--completions will be suggested Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Program Selection

Algorithm

- (●) blastp (protein-protein BLAST)
- ( ) PSI-BLAST (Position-Specific Iterated BLAST)

Choose a BLAST algorithm

BLAST Search database ClusteredNR using Blastp (protein-protein BLAST)

Show results in a new window

+ Algorithm parameters

Clusters producing significant alignments Download Select columns Show 100

select all *100 clusters selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
⊞17 member(s), 8 organism(s)	beech family	hypothetical protein ACB092_11G070700 [Castanea dentata]	439	439	100%	3e-145	97.61%	1001	KAL4598604.1
⊞2 member(s), 2 organism(s)	beech family	hypothetical protein RGQ29_018155 [Quercus rubra]	431	431	100%	8e-145	96.65%	761	KAK4594360.1
⊞1 member(s), 1 organism(s)	Japanese beech	ubiquitin carboxyl-terminal hydrolase 15 [Fagus crenata]	419	419	100%	1e-137	92.34%	962	GMY33491.1
⊞1 member(s), 1 organism(s)	eudicots	hypothetical protein FH972_019460 [Carpinus fangiana]	409	409	100%	7e-135	90.91%	880	KAE8124593.1
⊞2 member(s), 2 organism(s)	birch family	ubiquitin carboxyl-terminal hydrolase 15 [Alnus glutinosa]	402	402	100%	3e-131	89.47%	952	XP_062168959.1
⊞2 member(s), 2 organism(s)	eudicots	ubiquitin carboxyl-terminal hydrolase 15 isoform X1 [Prunus yedoen...	384	384	100%	7e-128	85.17%	627	PQP99729.1
⊞8 member(s), 2 organism(s)	eudicots	ubiquitin carboxyl-terminal hydrolase 15-like [Pyrus ussuriensis x Pyr...	384	384	100%	5e-127	85.65%	696	KAB2636749.1
⊞1 member(s), 1 organism(s)	apple	hypothetical protein ACFX2G_036545 [Malus domestica]	386	386	100%	1e-126	85.65%	792	KAM1244409.1
⊞1 member(s), 1 organism(s)	pistachio	ubiquitin carboxyl-terminal hydrolase 15 [Pistacia vera]	390	390	100%	1e-126	85.17%	986	XP_031285705.1
⊞2 member(s), 1 organism(s)	shan jing zi	unnamed protein product [Malus baccata var. baccata]	385	385	100%	2e-126	85.65%	782	CAN6688914.1
⊞1 member(s), 1 organism(s)	eudicots	ubiquitin carboxyl-terminal hydrolase 15-like isoform X3 [Actinidia eri...	385	385	100%	3e-126	83.73%	827	XP_057506896.1
⊞1 member(s), 1 organism(s)	eudicots	unnamed protein product [Rubus chamaemorus]	384	384	100%	4e-126	85.65%	771	CAO1787437.1

hypothetical protein ACB092_11G070700 [Castanea dentata]

Sequence ID: KAL4598604.1 **Length:** 1001 **Number of Matches:** 1

Range 1: 454 to 662 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
439 bits(1129)	3e-145	Compositional matrix adjust.	204/209(98%)	208/209(99%)	0/209(0%)

```
Query  1    FFQCEVFDLSPRGLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKGWCLMCEL  60
            FFQCEVFDLSPRGLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKGWCLMCEL
Sbjct  454  FFQCEVFDLSPRGLLNCGNSCYANAVLQCLTCTKPLITYLIHRSHSRACCVKGWCLMCEL  513

Query  61   EQHVVMLRESGGPLSLSRILMHIRSINFQIGDGSQEDAHDFLRLIIASMQSICLEGLGGE  120
            EQHVVMLRESGGPLSLSRILMHIRSINFQIGDGSQEDAH+FLRLIIASMQSICLEGLGGE
Sbjct  514  EQHVVMLRESGGPLSLSRILMHIRSINFQIGDGSQEDAHEFLRLIIASMQSICLEGLGGE  573

Query  121  KKVDPILQETTFIQHIFGGRLRSKVKCMICHQESERYENMMDLTLEIYGWVESLEDALTQ  180
             KVDPILQETTFIQHIFGGRLRSKVKCMICHQESERYEN+MDLTLEIYGWVESLEDALTQ
Sbjct  574  NKVDPILQETTFIQHIFGGRLRSKVKCMICHQESERYENIMDLTLEIYGWVESLEDALTQ  633

Query  181  FTATEDLDGENMYRCGRCATYVRAKKELS  209
            FTATEDLDGENMYRCGRCATYVRA+K+LS
Sbjct  634  FTATEDLDGENMYRCGRCATYVRARKQLS  662
```

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width.

Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Relabelled:

Human ubiquitin carboxyl-terminal hydrolase 36 isoform 2 NP_001372103.1 -> Human Deubiquitinase -> Human DUB
Quercus robur sequence (from BLAST) -> English Oak
Castanea dentata -> American chestnut
Quercus rubra -> Northern Red Oak
Fagus crenata -> Japanese beech
Carpinus fangiana -> monkeytail hornbeam
Alnus glutinosa -> common alder

Salix viminalis -> basket willow
Prunus yedoensis var. nudiflora -> Korean king cherry
Malus baccata var. baccata -> Siberian crabapple
Catharanthus roseus -> Cape periwinkle
Malus domestica -> Apple
Punica granatum -> Pomegranate
Pistacia vera -> Pistachio
Mangifera indica -> Mango
Ipomoea batatas -> Sweet potato

Alignment in SeaView

```
Human DUB            ---------- ---------- ---------- ---------- ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    MLEPREADIP VLFLVLVVLP LVAYILLGKW SEVTKKRERI SLLAQLAAEE AFKAEAMAVG
Northern red oak     MLEPREADIP VLLLVLVVLP LVAYILLGKW SEVTKKRERI SLLAQLAAEE AFKAEAMAVG
Japanese beech       MLEPREADIP VLFLVLVVLP LVAYILLGKW SEVTKKRERI SLLAQLAAEE ALTTEAMAVP
monkeytail hornbeam  MLEPRQADIP VLFLVLVVLP LVAYILLGKW SEATKKRERI SLLAQVAAEE ALRAETMAVA
common alder         MLEPRQADIP ALFLVLVVLP LVAYILLGKW SEAAKKRERI SLLAQVAAEE NLRAETMAAA
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   MLEPREADIP ALFLVFVVLP LVAYVLLGKW SETSKKTERV SLLAQLAAEE ALRVETMAIA
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                MLEPREADIP VLFLVFVVLP LVAYVLLGKW SDSSKKRDRV SLLAQLAAEE ALRAEAMAVA
Pomegranate          MLEPREADIP ILFLVLVVLP VVLYILLGKW SDAAKKKERV NLLAQLAAEE ALRVETMAVA
Pistachio            MLEPREADVP VLFLVLVVLP VVVYFLLGKW SETAKKRERI SLLAQLAAEE AFKAETMATA
Mango                MLEPREADVP LLFLVLVVLP VVAYFLLGKW SETTKKRERI SLLAQLAAEE AFEEEAMATA
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                    61
Human DUB            ---------- ---------- ---------- ---------- ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    NAI-PL--VS SSKSGLHVCA RCFAPAKTRC SRCKSIRYCS GKCQIIHWRQ AHKQECQQLE
Northern red oak     NAI-PL--VS SSKSGLHVCA RCFAPAKTRC SRCKSIRYCS GKCQIIHWRQ AHKQECQQLE
Japanese beech       NVI-PL--LS SSK---TVCA NCFAPATTRC SRCKSIRYCS GKCQIFHWRQ AHKQDCQQLE
monkeytail hornbeam  NTI-PL--IA SSKIGHHACA RCFAPATTRC SRCKSVRYCS GKCQIIHWRK AHKRECQQLE
common alder         NAI-PL--MA SSKSGLHVCA RCCAPATTRC ARCKSVRYCS GKCQIIHWRK AHKRECQQLE
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   AVIPPVPSSN PSKNGLQVCA RCYSTATTRC SRCKSIRYCS GKCQIIHWRE VHRQECLQLE
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                AVIPPVSSSS PSKNGLQVCA RCYSTATTRC SRCKSVRYCS GNCQIIHWRE VHRQECLQLE
Pomegranate          ATVVPHVVPP KPKSSLHECA QCLSPATTRC SRCKSVRYCS GKCQIIHWRQ VHQHECQPLD
Pistachio            SVI-PL--VA TSKNVLHVCA RCSAPSTTRC SRCKSVRYCS GKCQIIHWRQ GHKQECQQLE
Mango                SVI-PI--VA TSKNVLHVCA RCSAPSTTRC SRCKSVRYCS GKCQIIHWRQ GHNQECQQLE
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                   121
Human DUB            ---------- ---------- ---------- ---------- ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    TTSSSSSSPK SVSVDESLHG RVLLKDYMNS QFLGYNN-DQ G-EKVSTDDL T--SLSFGVP
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Northern red oak     TTSSSSSSPK SVSVEESLHG RVLLNDYMNS QFLGYNN-DQ G-EKVSTDDL T--PLSFGVP
Japanese beech       ATSSSSSSPK SVSVEDSLHG RVLLNDCMNS HFLGYDI-NQ G-EKASTDDI T--SLSIGVP
monkeytail hornbeam  TNSS-SSSPK SLSIEESIHG RVLLNESMKS QFLGYNI-NL R-DKAPTDDI TYPSVSIGVS
common alder         NTSS-SSSPK SLSVEESIHE RVLLNESMKS QFLGYNI-NQ R-DKAPTDDI TYPSISIGVS
basket willow        ---------- ---------- ---------- ---------- ----MLVLHS IMSITSTGAS
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   PTSS-SSSPK AVSFGESFHE KFLPNDGFNS QYFGCKMEHI VAEQAPPDNI IYPSTSTGAP
Cape periwinkle      ---------- ---------- ---------- ---------- MLENFPPENI DNPLAGSLPS
Apple                PTCS-SSSPK AVSFGESFHE KFLPNDSINS QYFGSKMEQI LAEQAPADNI IYPSTSTGVP
Pomegranate          SNSEDSMDYQ S--------- ---------- --HGCNFQEA TVEKLPSENH LA--------
Pistachio            KTSSCSS--- --PLAESIQD SVLLNESLNS QLSGYYS-KT GMERAPFDDM VHPSIGRGAF
Mango                KTSSCSS--- --PLAESIQD SVLLYENLSS QLSGHYS-KA GMERAPLDDM IHPSICRGAY
Sweet potato         ---------- ---------- ---------- ---------- -------MKV M--GFYISVV

                   181
Human DUB            ---------- ---------- ---------- ---------- ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    AVVDCS-AIG NSQIPILER- TVNKRVSHKP QNEMLRGEDK DVIDSCEDAS KSRATHL---
Northern red oak     AVVDCS-AIG NSQIPILER- TVNKRVSHKP QNEMLRGEDK DVIDSCEDAS KSRATHL---
Japanese beech       AAADCS-VIS TSQIPILER- TMDKRVSHKS HGEMLRREDE DVIESCEEAS KSWATHST--
monkeytail hornbeam  AAGDCS-AVG TSQVPILERR TTDKRASRKS HSEVLRREDK DVIDAYEEAS KSSTTHS---
common alder         AAGDCS-AVG TSQIPMLERR TMGKRVSRKS HSEVLRREDK DVIDTYEEAS KSGTTHS---
basket willow        ASVNCP-ATD TSQEAMLHRR STNKQVSCKS NKEMLRRQEV AVFDSSQETA GIRPNNLT--
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   VMLDCASSVD TSQVSMPERR SVDKWVSRKS HREL-RKKVG VAFDSSEETS FGWTTHSG--
Cape periwinkle      GTADC--ITD SSGAPMAEKR SIDRRHSRKG AS---RRADG LLSECSGQVS E---------
Apple                DMLDCS-SVD TSQVSMPERR SVDKRVSRKS QREL-RKKVG VAFDSSEETS FGWTTHLE--
Pomegranate          ---SYV-SDE ASQAHVPARR PREKRISRKC NREMLRKEDG VVSVGFEEAR GNVGSTDNIM
Pistachio            AAMHCS-AVD SSHISMLER- TADKRACRKS NRESLRREDA TVFDSDEESP GSCTTNPN--
Mango                AATHCS-AVD SFHISMLER- TADKRVCCKS NKEPLRREDT TVFDSDKGSL GSCSTNPN--
Sweet potato         PL-------- ---------- ---------- ---------- ---------- ----------

                   241
Human DUB            ---------- ---------- ---------- ---------- ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    TSDFISSKEA FTRHRS-RKS DSVVSDVENL RKHNVNSSNV ---GQAASRG TMHGSDKYIT
Northern red oak     TSDFISSKEA FTRHRS-RKS DSVVSEVENL RKHNVNSSNV ---GQAASGG TMHGSDKYIT
Japanese beech       SSDFISSKET FIRNRS-RYS DSVVSEVENP RKHKVNSSNV YIHGQSASRD TMHEIDKYKT
monkeytail hornbeam  TSNFISSKEV LIRHR----- ---------- ---------- ---------- ----------
common alder         TSNFISSKEA LIRHKS-RNS DSVVSKVETP RNPSVICSNV NIHGQAAS-S TMHESDKYKN
basket willow        -SG------- ---------- ---------- ---------S ISNG-QDTTS AMDESHKYSI
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   SSNVNSSKEV FMEHKL-RNC DSHLAEEENA MMQNFNISDS YMKGQATSRN TVREIDKSQS
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                SSNVNSSKE- -----L-RNY DTQLAGEGNP MVQNVNISDS YMKGQATSRN TVHENDKSQG
Pomegranate          SPNACAQKEI LTKHKIMQNK EARVLEEGTP RKSKVRNSST YINGNATSVD AIHENHKCET
Pistachio            SVIDVPPKEA PMRHKS-RTS DFVVSAEETR KRHNVNGSNV HTHGPA--RS TGHENHYYQS
Mango                SSINVSPKES PLKHKS-RTS DVLVSAEESW KRHSVNGSNV HTQGPA--RS TIHENHYYQS
Sweet potato         SSCFICCL-- ---LQL-RES DSMLYEQYS- ---------- VPGGDADNKS TMQ--NSYMS

                   301
Human DUB            ---------- ---------- ---------- --XM------ ---------- ----------
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    PPQNTLEPRK SCGFSSS--S CLTRNGTNAH --QME----Y DQDEGDLLKW ENSTNA-EKI
Northern red oak     PPQNTFEPRK SCGFSSS--S CLTRNGTNAH --QME----Y DQDEGDLLKW ENSTNA-EKI
```

```
Japanese beech       PLRNSFEPRS SCGFSSS--S CLTRNGTSAH --QME----S DQYDGDLLKW GNTTNV-DTV
monkeytail hornbeam  ---------- ---------- ------VNAH GIGME----C DKYGGDLLEW ENTTNV-NTV
common alder         QHGTIVEPRN SCGFSRS--S HWARNGTDAH EIGME----C DKYGGDSLEW ENTTNV-NTV
basket willow        QHQNVSKSRS NHASSSS--M NAGKYGTNAH EIGTNIADFC NAEA------ --SFNG-EMT
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   QSGYIYESGS NSGSTSS--S YSSKSGIDVH ENGLD----F IPNGGNYLKE ERLSNY-ETA
Cape periwinkle      --SGVTSARL SCDAPLKEAS V-------RH KLR------- ---------- -------ESQ
Apple                QNGNIYESRS NSGLTSS--A YSSKSGIDVH ENGLD----F IPNGGKSLKG EILSND-ETT
Pomegranate          QQDSSCSSS- ---------- -----GTQKN IHQTPPPPYI KPDYGSSVQS GGAASG----
Pistachio            QNGNISEPRS SSGFPVA--P YSAKYGTNAH ENETD----I VP-------K G------NDE
Mango                QLGNVSESRG SSGFPVA--P YSAKYGTNDH KNETD----I VP-------G G------NDK
Sweet potato         QRQSENDPWT KSYYASSDAP NLESHETNAW --ETQ----V DTSNGDNILS GGFTSQDMTA

                   361
Human DUB            -----PIVDK LKEALKPGRK DSADDGELGK LLASSAKKVL LQKIEFEPAS KSFSYQLEA-
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    NFNGSYEMTA TNGNVNAKSG SQALQEKISK LPKSTMK--- -------VSG EKYCSELERK
Northern red oak     NFNSSYDMTA TNGNVNAKSG SQALQAKISK LPKSTMK--- -------VSG EQYCSELERK
Japanese beech       NINCSSEMTA TNGSVKANSG SQSIQMKISK FPKSTSK--- -------VSG EQYHSEVERK
monkeytail hornbeam  NFNCSSEMTM KNGSIKAKGG SNSLRTKISK LPKSAMQ--- -------VSG EQCCSVIEKK
common alder         NFNCSPEMTS NNGSIKAKGG SHSLRTKISK LPKSTMQ--- -------VSG EHCCSGIQRK
basket willow        GPKCSYETPV MNENVKANSA SHPWETSLL- IIKSKMK--- -------VSG NQSYSEIDGK
Korean king cherry   ---------- ---------- ---------- -----ME--- -------VSG EQYFSEIEMQ
Siberian crabapple   EFKC-AEMTK MKTDVKAKRA PYSLGTKLSK LPKSTME--- -------VSG EQCYSEREGQ
Cape periwinkle      TLPASPEID- ---------- ISAGGTKLHK SPRSTGN--- -------AS- REQCPD----
Apple                DFKC-AEMTT MKASIKVKRA PYSLGRKVSK LPNSTME--- -------VSG EQCYSEIERQ
Pomegranate          ---------- ------IKS- ----GTRISK --KQSGK--- -------ASQ EQGCIEVD--
Pistachio            MVDCYSDMTA LNGILKVKNA LNLRGTKICK SAKSSTKLVG EQPC-MTVVG EQLCLEKERK
Mango                VVDCYSDMTA VNGNINVKKA LNPRGTKSCK LAKSSTKLVG ---------- EQLCSEKERK
Sweet potato         GMDYSCEM-- ---------- -----GATRK INKSKPK--- -------VSR DEMCSDQDGE

                   421
Human DUB            ------L-KS KYX---XVLL NPKTEGASRH ----KSGDDP P--ARRQGSE HTYESCGDGV
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    GHIANES--- NVARKRDIKA AQGSNGVERM GIMKMVGLRK STKLTRQDA- ---PINNGKR
Northern red oak     GHIANES--- NVARMRDIKA AQGSNGVERM GIMKMVGLRK STKLTRQDA- ---PINNGKR
Japanese beech       GKISNESSQT EVARMRDIKA AQGSNGVARM GIMKMVGLRK STKHSRQEA- ---PVTNDKR
monkeytail hornbeam  GQISNQ---S RVARKSDTVA AQEGNADVKA GIKKMVGLRK SIKLTGQGA- ---PIIPGKR
common alder         GQISNQLSQS EVARMSDIVA AQEGNADAKV GIKKMVGLRK SIKLTGQGA- ---PVIHAKR
basket willow        GQLTDES--- KVARMSNANP APGSNGATSI DIMKMMGLRK SAKLARQDT- ---AENN-IV
Korean king cherry   GHIAEDS--- KVARMRDT-T TQGSNGIANL RIMKMMGLKK PKTISREDV- ---PEVNGYR
Siberian crabapple   EQIAEVS--- KVTRMRDT-T AQGSNGIANV RIMKMMGLKK PKNFTRQEV- ---SELNGYQ
Cape periwinkle      ------LDLK EITRTKDAIP LQGKDSVSGL GIMKMIGLVK PSKLDRQEN- ---KEANSNR
Apple                GHIAEDS--- KVTRMRDT-T AQGSNGIANS RIMKMMDLKK PKNLTRQEV- ---PEVNGYR
Pomegranate          ---------- -----KRAQA VEGSNAIANM GIMKMMGLRK SVKVVRQDA- ---WELNGEG
Pistachio            GLIDDES--- KVPRVRENIT AQRSSGVSKM GIMKMMGLKK SSKLRQDAP- ---ELWL--G
Mango                GLTEDEL--- KFPRVRENIT AQRSSGVSKM GIMKMMGLKK SSKLRQDAP- ---ELWL--E
Sweet potato         GQKGDEL--- -----RNAKI KESIPSNGRN GIMKMFGLMK SSRPDRQAH- ---SVVKTDK

                   481
Human DUB            PAPQKVLFPT ERLSLRWERV FRVXXXGAGL HNLGNTCFLN ATIQCLTYTP PLANYLLSKE
English Oak          ---------- -----FFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLITYLIHRS
American Chestnut    NKKLKMLFPY EEFVKFFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLITYLIHRS
Northern red oak     NKKLKMLFPY EEFVKFFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLITYLIHRS
Japanese beech       NEKLKMLFPY EEFLKFFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLVTFLLRRS
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monkeytail hornbeam  PKKSKMLFPY EEFVEFFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLIIYLLRRS
common alder         HKKSKMLFPY EEFVEFFQC- GVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLIIYLLRRS
basket willow        HKTTRMLFPY EEFVKIFNC- EVINLTPRGL VNCGNSCYAN AVLQCLTCTK PLIIFLLHRS
Korean king cherry   HKTKKVLFPY DEFVKYFQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLVMYLLRRS
Siberian crabapple   QKT-KVLFPY DEFVKYLQC- EVFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLIIYLLRRS
Cape periwinkle      HKKVKMLFPY EEYVKFFQY- DGFSVSPRGL LNCGNSCYAN AVLQCLSCTK PLINFLLRRS
Apple                QKTKKVLFPY DEFVKYFQC- EFFDLSPRGL LNCGNSCYAN AVLQCLTCTK PLIIYLLRRS
Pomegranate          QKKTKMLFPY EEFVKFFQY- EVFDLLPRGL INCGNSCYAN AVLQCLTCTR PLTIYLLRRS
Pistachio            KKTMKMLFPY EEFVKFFQC- EVFDLSPRGL VNCGNSCYAN AVLQCLTCTK PLVIYLLHRS
Mango                KKKMKVLFPY EEFVKFFHC- EVFVLSPRGL VNCGNSCYAN AVLQCLTCTK PLVVYWLHRS
Sweet potato         HKKLKMLFPY EEFVKYFQY- EVLNVSPRGL INCGNSCYAN AVLQCLTFTK PLTVYLLHRS

                   541
Human DUB            HARSCHQGSF CMLCVMQNHI VQAFANXXXS GNAIKPVSFI RDLKKIARHF RFGNQEDAHE
English Oak          HSRACCVKGW CLMCELEQHV VM----LRES GGPLSLSRIL MHIRSINFQI GDGSQEDAHD
American Chestnut    HSRACCVKGW CLMCELEQHV VM----LRES GGPLSLSRIL MHIRSINFQI GDGSQEDAHE
Northern red oak     HSRACCVKGW CLMCELEQHL VM----LRES GGPLSLSRIL MHIRSINFQI GDGSQEDAHE
Japanese beech       HSRACCVKDW CLMCELEQHV VM----LRES GGPLSLSRIL SHIRSINFQI GDGSQEDAHE
monkeytail hornbeam  HSRACCKKGW CLMCELEQHV MM----LRES GGPLSLSKIL FHIRSINCQI GDGSQEDAHE
common alder         HSRACCRKDW CLMCELEQHV MM----LRER GGPLSPSRIL LHIRSINCQI GDGSQEDAHE
basket willow        HSRACCGKDW CLMCELEQHV MM----LRES GGPLSPSRIL LHMRKINCQI GDGSQEDAHE
Korean king cherry   HSSACCGKDW CLMCELEQHV MM----LRES GGPLSPSRIL FHMRSINCQI GDGSQEDAHE
Siberian crabapple   HSRACCGKDW CLMCELEQHV MM----LRES GGPLSPSRIL FHMRSINCQI GDGSQEDAHE
Cape periwinkle      HSRACCVRDW CLMCELEQHV MM----LRES GGPLSPSRIL SHMRSINCQI GDGSQEDAHE
Apple                HSRACCGKDW CLMCELEQHV MM----LRES GGPLSPSRIL FHMRSINCQI GDGSQEDAHE
Pomegranate          HSRACCGKDW CLMCELEQHV TM----LRET GAPLSPSRIL SHMRSINRQI GDGSQEDAHE
Pistachio            HSRACCGKDW CLMCELEQHV MM----LRES GGPLSPGRIL LHMRSINCQI GDGSQEDAHE
Mango                HSRACCGKDW CLMCELEQHV IM----LRES GGPLSPGRIL LHMRSINCQI GDGSQEDAHE
Sweet potato         HSRTCCGRDW CLMCELEQHV MM----LRES GEPLSPNRFL LRMQSINSQI GNGSQEDAHE

                   601
Human DUB            FLRYTIDAMQ KACLNGCAKL DRQTQATTLX XXVHQIFGGY LRSRVKCSVC KSVSDTYDPY
English Oak          FLRLIIASMQ SICLEGLGGE KKVDPILQET TFIQHIFGGR LRSKVKCMIC HQESERYENM
American Chestnut    FLRLIIASMQ SICLEGLGGE NKVDPILQET TFIQHIFGGR LRSKVKCMIC HQESERYENI
Northern red oak     FLRLIIASMQ SICLEGLGGE NKVDPILQET TFIQHIFGGR LRSKVKCMIC HQESERYENI
Japanese beech       FLRLLIASMQ SICLEGLGGE NKVDPILQET TFIQHIFGGR LRSKVKCMMC HHESERYENF
monkeytail hornbeam  FLRLLIASMQ SICLEGLGGE NKVDPILQET TFIQHIFGGR LRSKVKCMIC HHESERYENI
common alder         FLRLLIASMQ SICLEGLGGE NKVDPILQET TFIQHIFGGR LRSKVKCMIC HHESERYENI
basket willow        FLRLLIASMQ SICLERLGGE DKVDPRLQET TFIQHTFGGR LRSKVKCLRC HHESERYENI
Korean king cherry   FLRLLITSMQ SICLEGLGGE NKVDPRLQET TFIQHTFGGH LRSKVECLRC HHESERYENI
Siberian crabapple   FLRLLVTSMQ SICLEGLGGE SKVDPRLQET TFIQHTFGGH LRSKVKCLRC HHESERYENI
Cape periwinkle      FLRLLVASMQ SICLESLGGE NVVDPRLQET TFIQHTFGGR LRSKVKCLRC HHESQRSENI
Apple                FLRLLVTSMQ SICLEGLGGE NKVDPRLQET TFIQHTFGGH LRSKVKCLRC HHESERYENI
Pomegranate          FLRLLVASMQ SICLEGLGGE RKVDPKLQET TFIQHTFGGR LRSRVKCLTC HHESERYENI
Pistachio            FLRFLVASMQ SICLEGLGGE SRVDPRLQET TFIQHTFGGR LMSKVKCLRC HHESERYENI
Mango                FLRFLIASMQ SICLEGLGGE SRVDPRLQET TFIQHTFGGR LLSKVKCLRC HHESERYENI
Sweet potato         FLRLLVASMQ SICLEGLGGE KAVEPRLQET TFIQHIFGGR LRSKVKCLRC HHESECYENI

                   661
Human DUB            LDVALEIRQA -ANIVRALEL FVKADVLSGE NAYXXXMCAK CKKKVPASKR FTIHRTSNVL
English Oak          MDLTLEIYGW VESLEDALTQ FTATEDLDGE NMYR---CGR CATYVRAKKE LSTYPESEQX
American Chestnut    MDLTLEIYGW VESLEDALTQ FTATEDLDGE NMYR---CGR CATYVRARKQ LSIHEAPNIL
Northern red oak     MDLTLEIYGW VESLEDALTQ FTAIEDLDGE NKYR---CGR CATYVRARKE LSIHEAPNIL
Japanese beech       MDLTLEIFGW VESLEDALTQ FTATEDLDGE NMYR---CGR CATYVQARKQ LSIHEAPNIL
monkeytail hornbeam  MDLTLEIFGW VESLEDALTQ FTTAEDLDGE NMYR---CAR CATYVRARKQ LSIHEAPNIL
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common alder         MDLTLEIFGW VESLEDALTQ FTTAEDLDGE NMYR---CAR CATYVRARKQ LSIHEAPNIL
basket willow        MDLTLEIFGW VESLEDALTQ FTTPEELDGE NMYR---CGR CAAYVRARKQ LSIHEAPNIL
Korean king cherry   MDLTLEIFGW VESLEDALTQ FTTPEDLDGE NMYR---CGR CAAYVRARKQ LSIHEAPNIL
Siberian crabapple   MDLTLEIFGW VESLEDALTQ FTTPEDLDGE NMYR---CGR CAAYVRARKQ LSIHEAPNIL
Cape periwinkle      MDLTLEIFGW VESLEDALTQ FTSPEDLDGE NMYR---CGR CATYVRARKQ LSIQEAPNIL
Apple                MDLTLEIFGW VESLEDALTQ FTTPEDLDGE NMYR---CGR CAAYVRARKQ LSIHEAPNIL
Pomegranate          MDLTLEIYGW VESLEDALTQ FTTPEDLDGE NMYR---CGR CATYVQARKQ LSVHEAPNIL
Pistachio            MDLTLEIFGW VESLEDALTQ FTTAEDLDGE NMYR---CGR CASYVRARKQ LSIHEAPNIL
Mango                MDLTLEIFGW VESLEDALTQ FTTAEDLDGE NMYR---CGR CASYVRARKQ LSIHEAPNIL
Sweet potato         MDLTLEIYGW VESLEDALTQ FTSPEDLDGE NMYR---CGS CSSYVRAQKQ LSIQEAPNIL

                   721
Human DUB            TLSLKRFANF SGGKITKDVG YPEFLNIRPY MSQNNGXXXD PVMYGLYAVL VHSGY--SCH
English Oak          ENCEX----R PASTEITWIT HCMDV----- ---------- ---------- ----------
American Chestnut    TIVLKRFQEG RYGKISKWIT FPDMLDMIPF MTG---TEDI PPLYMLYAVV VHLDTLNASF
Northern red oak     TIVLKRFQEG RYGKISKWIT FPDMLDMIPF MTG---TEDI PPLYLLYAVV VHLDTLNASF
Japanese beech       TIVLKRFQEG RYGKINKWIT FPDMLDMIPF MTG---TRDI PPLYMLYAVV VHLDTLNASF
monkeytail hornbeam  TIVLKRFQEG RYGKLNKWIT FPDMLDMIPF MTG---TGDM PPLYMLYAVV VHLDSLNASF
common alder         TIVLKRFQEG RYGKINKWIT FPDMLDMIPF MTG---TGDI PPLYMLYAVV VHLDSLNASF
basket willow        TIVLKRFQEG KYGKINKCIT FPDMLDMIPF MTG---SGDV -LLYMLYAVV VHLDTLNASF
Korean king cherry   TIVLKRFQEG KYGKINKCIT FPDMLDMVPF MTG---KGDI PPLYLLYAVV VHLDTQNASF
Siberian crabapple   TIVLKRFQEG NYGKINKCIT FPDMLDMIPF MTG---TGDI PPLYLLYGVV VHLDTQNASF
Cape periwinkle      TIVLKRFQEG SYGKINKCIS FPDMLDMVPF MTG---TDDV PPLYMLYGVV VHLDTLNASF
Apple                TIVLKRFQEG KYGKINKCIT FPDMLDMIPF MTG---TGDI PPLYLLYGVV VHLDTQNASF
Pomegranate          TIVLKRFQEG RYGKINKCIT FPEMLDMIPY MTG---TGDM PPLYLLYAVV VHLDTLNASF
Pistachio            TIVLKRFQEG RYGKINKCIT FPEMLDMIPF MTG---TGDM PPLYMLYGVV VHLDTQNASF
Mango                TIVLKRFQEG RYGKINKCIT FPAMLDMIPF MTG---TGDM PPLYMLYGVV VHLDTQNASF
Sweet potato         TIVLKRFQGG CYGKINKCIT FPEMLDMIPY MTG---TDDV PPLYMLYAVV VHLDTLNASF

                   781
Human DUB            AGHYYCYVKA SNGQWYQMND SLVHSSNVKV VLNQ------ QAYVLFYXXX LRIPGSKKSP
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    SGHYVSYVKD MRGNWFRIDD TEVQPVLMSQ VMSE------ GAYILFYMRS CPRPQ-RTYI
Northern red oak     SGHYVSYVKD MRGNWFRIDD TEVNFSPYK- ---------- ----CYGIR- ----------
Japanese beech       SGHYVSYVKD MCGNWFRIDD TEVQPVPMSQ VMSE------ GAYILFYMRS CPRPQ-RTYI
monkeytail hornbeam  SGHYVSYVKD LNGNWFRIDD TEVQPVPMSQ VMSE------ GAYILFYTRS CPRPQ-R--T
common alder         SGHYVSYVKD MNGNWFRIDD TEVQPVPMSQ VMSE------ GAYILFYTRS CPRPQ-RSYT
basket willow        SGHYVAYVKD LRGSWFKIDD TEVHPVSMSQ VMLE------ GAYILFYKRS CTRPQ-KLSR
Korean king cherry   SGHYVAYVKD MRGNWFRIDD TQVQPVSMSQ VMME------ GAYILFYMRS CPRPQ-RAFT
Siberian crabapple   SGHYVAYVKD MRGDWFRIDD TEVHPVSMSQ VMME------ GAYILFYMRY V---------
Cape periwinkle      SGHYISYVKD MRGNWFRIDD TEVQPVETSQ VMSE------ GAYILFYVRS SPRPG-RPCI
Apple                SGHYVAYVKD MRGDWFRIDD TEVNAFSNQH FSARVWGFCD NGLNLFSLNT NKMEVQSEYT
Pomegranate          SGHYVCYVRN LQGSWYRIDD TEVQPVLMSQ VMSE------ GAYILFYMRS SPRPH-KAYL
Pistachio            SGHYVSYIKD LQGSWFRVDD TEVHPVPMSQ VMSE------ GAYILFYMRS CPRPLPRASL
Mango                SGHYVSYIKD MEGNWFRIDD TEVHPVPMSQ VMSE------ GAYILFYMRS CPRPPPRTTS
Sweet potato         SGHYISYVKD MQGNWFRIDD TKVKFFIKLY C--------- -----FGFSN CRGKI-QT--

                   841
Human DUB            EGLISRTGSS SLPGRPSVIP DHSKKNIGNG IISSPLTGKR QDSGTMKKPH XXXTTEEIGV
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    EETIQQQVPD S--ARNCISK AQKLARTSQS RPSS------ ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       EKAIQQQVPD S--ARNCTSK AQKLARPGQS RPSS------ ---------- ----------
monkeytail hornbeam  DKAFQQQVPE S--ARNFIAK ASKPSRPVQS RLSS------ ---------- ----------
common alder         DKTFQQQVPE S--ARNCIAK AQKPSRPVQS RPSS------ ---------- ----------
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basket willow        EKPIRQQVPV S--SRHRTSR TQKPSRQGQS KCSS------ ---------- ----------
Korean king cherry   GKSIRQQVPN V--ENHCMSK AQK-SRPGQS KHSS------ ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      QKTSSEQASA V--PKQCPSR SQKQSRSEEC RLNH------ ---------- ----------
Apple                C--------- ---------- ---------- ---------- ---------- ----------
Pomegranate          GKSERETDPG S--LRHCALR SQKHSRPRQS KSSD------ ---------- ----------
Pistachio            GKAMRQPVQA S--ARHFTPK TQKSSRQGQN KASS------ ---------- ----------
Mango                GKAMRQPVPA S--ASHCTPK TQMSSRQGPN KASS------ ---------- ----------
Sweet potato         --KLXXTYPE SEQXENCEXR ---------- -PAS------ ---------- ----------

                   901
Human DUB            PISRNGSTLG LKSQNGCIPP KLPSGSPSPK LSQTPTHM-- PTILDDPGKK VKKPAXXX--
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- -----QSVGL Q-DSTNPRPD MPPRSTNHTS NGILKSA-SR NVQPVMEAYA
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- -----QYVGL E-PTTNSRPD MPRST----- NHTLKIA-SR NVLPVMEAYA
monkeytail hornbeam  ---------- -----QYAGL E-PSTDTRPD IASSSTNHT- --SIRSA-SR NVLPPMEAYA
common alder         ---------- -----QYVGL E-PSTDPRPD IASSSTNPT- --SFRSA-SR NVLPPMEAYA
basket willow        ---------- -----HCVGP E-VSWDIKPE NGSVPANY-A NGIPRRCAGK NI--------
Korean king cherry   ---------- -----QFVSP VHLPDDIRPE IANSFANSTS SDIFRSS-NG NVFPMTETYG
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- -----Q---- ---PSD--HK WGMNFANYSS TGSLRTA-NG SRTPITETRS
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- -----HLVVP P--------E PFSHSTKLAS NGVVRVTENG NIWPVVEKYS
Pistachio            ---------- -----HVVGP E-FSSDVRPE IVSGFTEYNS NGILRRASNK STLPVMERYT
Mango                ---------- -----DFVLR E-FSSDLRPE IVSTVTEHNF NGIRSRASNK STLPMMERYV
Sweet potato         ---------- -----TEITW I--------- ---------- ---------- ----------

                   961
Human DUB            --PPQHFSPR TA---QGLPG TSNSNSSRSG SQRQGSWDSR DV-------- VLSTSPKLLA
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    EPVCMDFSNA -TSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADTCNADT ISSIFNTLYA
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       EPGCMEFSNA -TSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADTCNIDT ISSIFNTLYA
monkeytail hornbeam  EPICMEFSDA -TSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADTCNIDS MSSIFNALYA
common alder         EPICMEFSDA -TSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADTFNIDS ISSIFNTLYA
basket willow        -TQVMDFSDA -TSSDRSIFT SSDEASFTTE S-TRDSFSTI DYADACNVDA FSSIFNDLYP
Korean king cherry   EPIGVEFSDA -TSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADAGN--- ISSIFNTLYA
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      QAM-NEFSDA TSSSDWSLFT SSDEASFTTE S-TRDSFSTV DYADASNVDP -FSIFNSLYA
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          -----EFSDA -TSSDWSLFT SSDDASFTTE S-TRDSFSTA DYSDNGNGDL ISSIFSNLYG
Pistachio            ESMSMEFSDA -TSSDWSLFT SSDEASFTTE S-ARDSFSTV DYSDTNGD-- PASIFKTGYA
Mango                EPTSMEFSDA -ASSDWSFFT SSDEASFTTE S-ARDSFSTV DYSDTNGD-- PVSIFMTGYA
Sweet potato         -THCMDV--- ---------- ---------- ---------- ---------- ----------

                  1021
Human DUB            T--------- ---------- ----ATANGH GLKGNDXXXE SAGLDRRGSS SS--SPEHSA
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    PEYS-RNSVS CRKVSNCRPH TRFVSEQKGH ILDSYLSTQP LDKVQKRVFS KQLSDPSSQF
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       PEYS-SNSVS CRKVSNCRPH TRFVSEQKGR ILDSYLSTQP LDKVQKRVFT K---------
monkeytail hornbeam  PEYS-RYSAS CRKFSSSRPH TGFLSEQKGH ILNS------ ---------- ----------
common alder         PEYS-RYSAS CRKFSSSRPH IGFVSEQKGH VLNS------ ---------- ----------
basket willow        PESSNRNTLC RRTFSYSRPQ TRFVSE-TGN VLDSYSTQSS GHEKERNPNR YVIHQPTSLR
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Korean king cherry   PEYS-RSSVS CRKFSNSRPN TRFVSQDKGL LLDSYLSTHP IDRVQNRNYS KQVSDSSTAR
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      PEYPPLKTIA CSVFATSKPQ TRYFSENMGA VWECYVPQVS FSSAEEA--- ---------L
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          QDYSPRNTVS CRLFSNSGAQ TRFASEEKGY VLDSYLSAQH NHRRRGLG-- ----------
Pistachio            ESAS-RNTVC CRTFSNSKPQ TRFSLEKKGY VLDSYLSTQS PHKLHRDEDF RQVDNSSAES
Mango                ESAS-RNTVC CRTFSNSKPQ TRFILEKKGY VLDS------ ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1081
Human DUB            SSDST-KAPQ TPRSGAAHLC DSQETNCSTA GHSKTPPSGA DSXXXKTVKL KSPVLSNTTT
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    PSDSNCNMFV KYGSNPIHGF ERTSDHCKL- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        AVPAPCL--- ---------- ---------- ---------- ---------- ----------
Korean king cherry   PPDGKCSTFV RYGSNPVHFL DRTSDHCKL- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      DTDNFCREDV TYWSDPRTGS PQNSVNCNV- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------T DRTSSHCKST S--------- ---------- ----------
Pistachio            SSDSNCDVFV NYRTSRF--- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1141
Human DUB            EPASTMSPPP AKKLALSAKK ASTLWRATGN DLRPPPPSPS SDLTHXXXPM KTSHPVVAST
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1201
Human DUB            WPVHRARAVS PAPQSSSRLQ PPFSPHPTLL SSTPKPPGTS EPRSCSSIXX XSTALPQVNE
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
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Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1261
Human DUB            DLVSLPHQLP EASEPPQSPS EKRKKTFVGE PQRLGSETRL PQHIREATAA PXXXHGKRKR
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1321
Human DUB            KKKKRPEDTA ASALQEGQTQ RQPGSPMYRR EGQAQLPAVR RQEDGTQPQV NGQQXXXVGC
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1381
Human DUB            VTDGHHASSR KRRRKGAEGL GEEGGLHQDP LRHSCSPMGD GDPEAMEESP RKKKKKKXXX
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
```
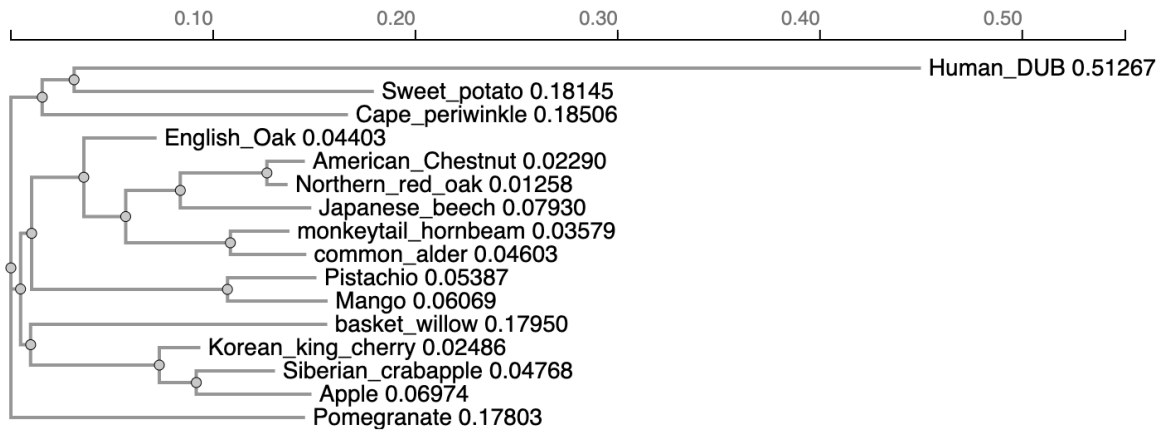

```
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1441
Human DUB            RKQETQRAVE EDGHLKCPRS AKPQDAVVPE SSSCAPSANG WCPGDRMGLS QAPPVSWNGE
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1501
Human DUB            XXXXRESDVV QELLKYSSDK AYGRKVLTWD GKMSAVSQDA IEDSRQARTE TVVDDWDEEF
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
Apple                ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate          ---------- ---------- ---------- ---------- ---------- ----------
Pistachio            ---------- ---------- ---------- ---------- ---------- ----------
Mango                ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato         ---------- ---------- ---------- ---------- ---------- ----------

                  1561
Human DUB            DRGKXXXXEK KIKKFKREKR RNFNAFQKLQ TRRNFWSVTH PAKAASLSYR RTYPESEQXE
English Oak          ---------- ---------- ---------- ---------- ---------- ----------
American Chestnut    ---------- ---------- ---------- ---------- ---------- ----------
Northern red oak     ---------- ---------- ---------- ---------- ---------- ----------
Japanese beech       ---------- ---------- ---------- ---------- ---------- ----------
monkeytail hornbeam  ---------- ---------- ---------- ---------- ---------- ----------
common alder         ---------- ---------- ---------- ---------- ---------- ----------
basket willow        ---------- ---------- ---------- ---------- ---------- ----------
Korean king cherry   ---------- ---------- ---------- ---------- ---------- ----------
Siberian crabapple   ---------- ---------- ---------- ---------- ---------- ----------
Cape periwinkle      ---------- ---------- ---------- ---------- ---------- ----------
```

```
Apple                 ---------- ---------- ---------- ---------- ---------- ----------
Pomegranate           ---------- ---------- ---------- ---------- ---------- ----------
Pistachio             ---------- ---------- ---------- ---------- ---------- ----------
Mango                 ---------- ---------- ---------- ---------- ---------- ----------
Sweet potato          ---------- ---------- ---------- ---------- ---------- ----------

                   1621
Human DUB             NCEXRPASTE ITWITHCMDV
English Oak           ---------- ----------
American Chestnut     ---------- ----------
Northern red oak      ---------- ----------
Japanese beech        ---------- ----------
monkeytail hornbeam   ---------- ----------
common alder          ---------- ----------
basket willow         ---------- ----------
Korean king cherry    ---------- ----------
Siberian crabapple    ---------- ----------
Cape periwinkle       ---------- ----------
Apple                 ---------- ----------
Pomegranate           ---------- ----------
Pistachio             ---------- ----------
Mango                 ---------- ----------
Sweet potato          ---------- ----------
```

[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

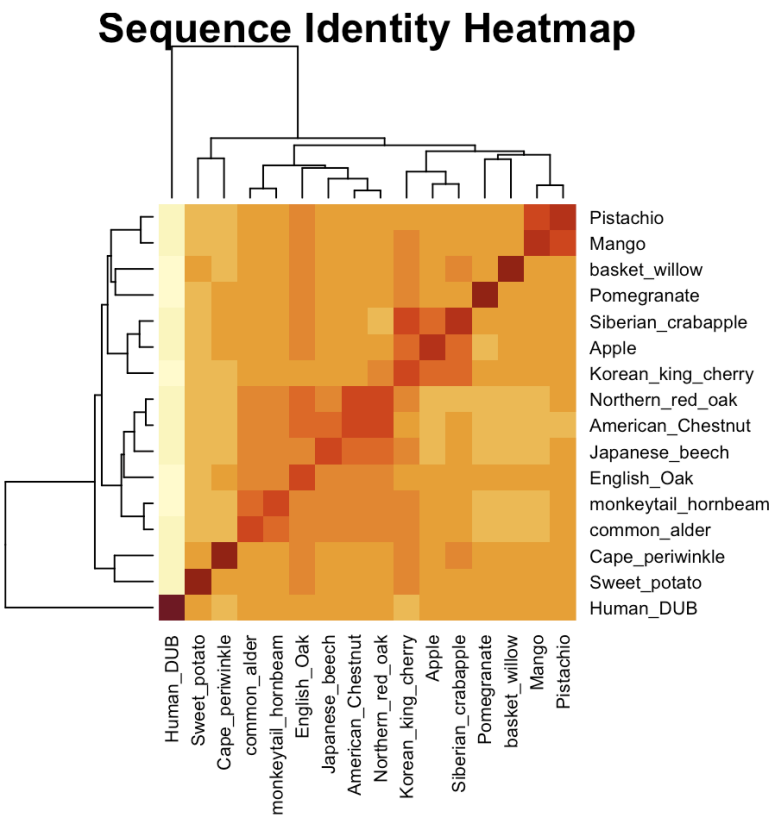
Phylogenetic tree generated with EBI Simple Phylogeny.
Parameters:
TREE FORMAT: Phylip
DISTANCE CORRECTION: OFF
EXCLUDE GAPS: OFF
CLUSTERING METHODS: Neighbour-joining
PERCENT IDENTITY MATRIX: OFF



[Q7] Generate a sequence identity based **heatmap** of your aligned sequences using R.

If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and "Save as" FASTA format for example). Read this FASTA format alignment into R with the help of functions in the **Bio3D package**. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

```
library(bio3d)
aln <- read.fasta(file = "yum.fst")
id <- seqidentity(aln)
heatmap(id,
        margins = c(8,8),        # increase margins
        cexRow = 0.8,            # reduce font size
        cexCol = 0.8,
        main = "Sequence Identity Heatmap")
```



[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 *unique* hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function consensus(). The Bio3D functions blast.pdb(), plot.blast() and pdb.annotate() are likely to be of most relevance for completing this task. Note that the results of blast.pdb() contain the hits PDB identifier (or pdb.id) as well as Evalue and identity. The results of pdb.annotate() contain the other annotation terms noted above.

Note that if your consensus sequence has lots of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by

calculating the row-wise maximum from your sequence identity matrix.

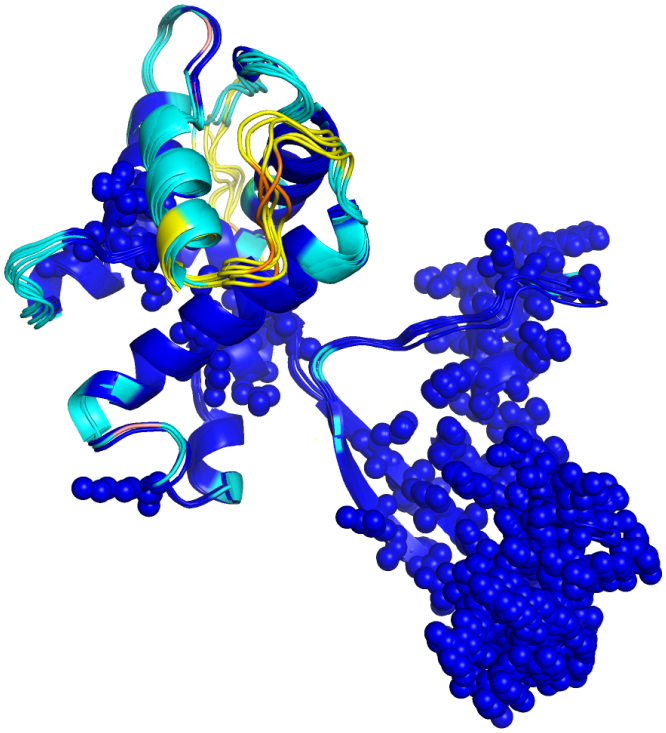
structureId	evalue	identity	experimentalTechnique	resolution	source
8BS3_A	3.06E-50	37.752	X-ray	2.2	Homo sapiens
6TBM_Q	7.10E-22	27.987	EM	20	Komagataella phaffii GS115
4FIP_A	1.58E-18	30.769	X-ray	2.686	Saccharomyces cerevisiae S288C

[Q9] Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence.

> Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for PFAM domain matches.

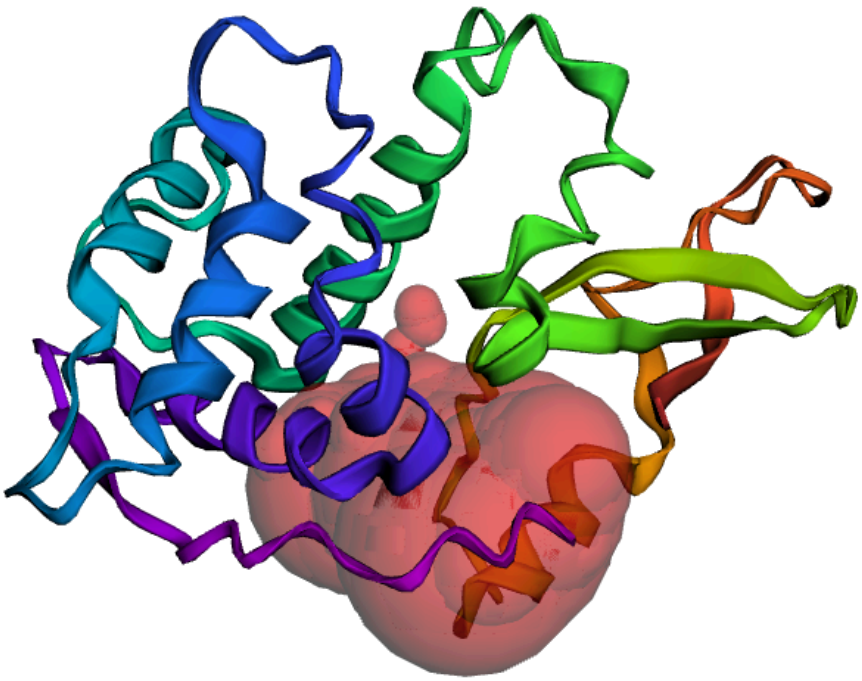
Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the **Mol* viewer** online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight *conserved residues* that are likely to be functional as **spacefill** and the protein as **cartoon** colored by local alpha fold *pLDDT quality score*. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that *pLDDT* score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).

Set at b>95



[Q10] (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (https://cfold.bme.uic.edu/castpfold/). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s **area** and **volume**.

Area: 497.935 volume: 1078.527



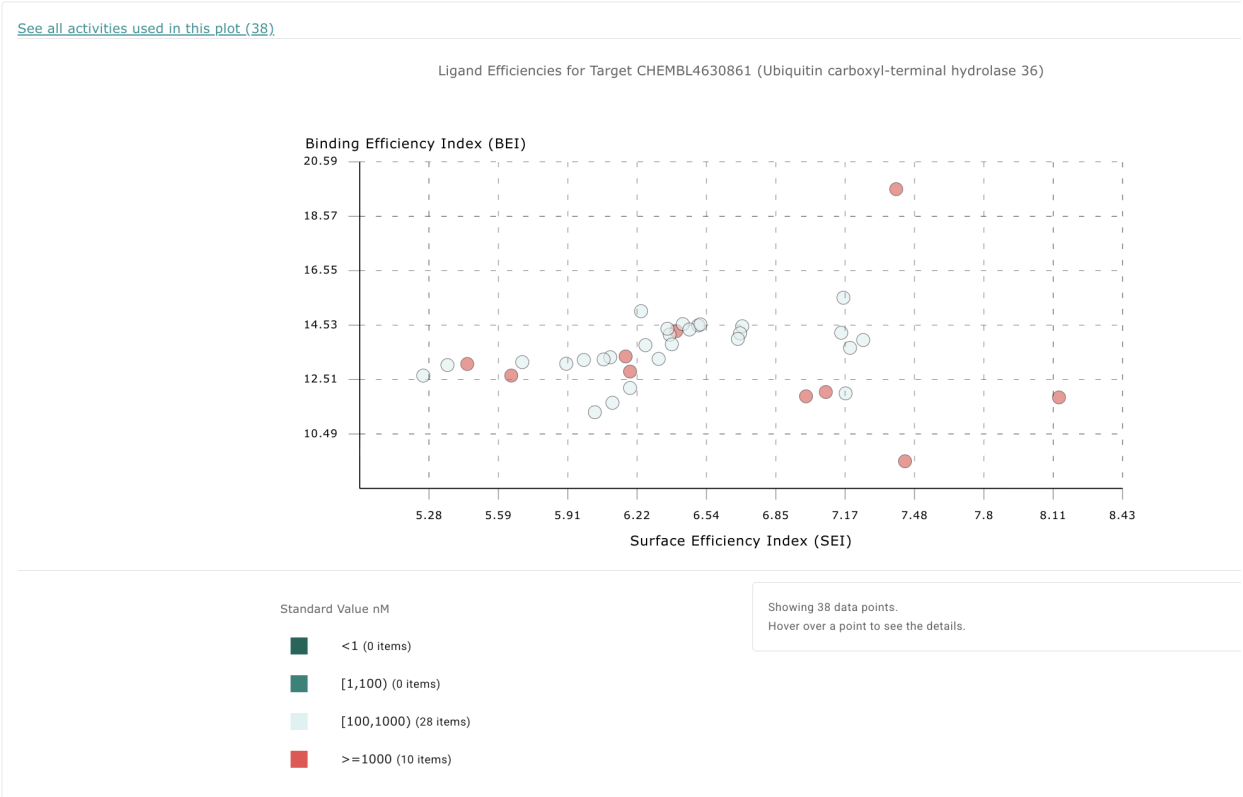
(ii) Perform a “Target” search of ChEMBEL (https://www.ebi.ac.uk/chembl/) with your novel sequence. Are there any **Target Associated Assays** and **ligand efficiency data** reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

Chembel job ID:
BIOLOGICAL_SEQUENCE_SEARCH-lsrDxFPwuaRAsMBzEk1PJfT6afqa0ysoj1MBZS4gNUo_eq_

Potential: Ubiquitin carboxyl-terminal hydrolase 36 CHEMBL4630861

ChEMBL ID	Assay Type	Description
CHEMBL5251166	B	Inhibition of USP36 (unknown origin) using ubiquitin rhodamine 110 as substrate at 1 to 10 uM by DUBprofiler fluorometric assay
CHEMBL4605988	B	Inhibition of human GST-tagged USP36 CD (81 to 461 residues) expressed in Escherichia coli assessed as cleavage of Ubiquitin-Rhodamine110-glycine to Ubiquitin and Rhodamine110-glycine using Ubiquitin-Rhodamine110-glycine as substrate by fluorescence based analysis
CHEMBL5393574	B	Inhibition of human USP36 (81 to 461 residues) using Ub-Rh110MP as substrate by fluorescence based assay
CHEMBL4674633	B	Inhibition of USP36 (unknown origin) at 10 uM relative to control
CHEMBL5112530	B	Inhibition of USP36 (unknown origin) using Ub-Rho measured after 0.5 to 3 hrs

Ligand Efficiencies



Molecule ChEMBL ID	Molecular Weight	AlogP	Ligand Efficiency BEI	Ligand Efficiency LE	Ligand Efficiency LLE	Ligand Efficiency SEI	Assay Description	Target Organism	Source Description

CHEMBL5 173239	426.83	3.99	12.05	0.23	1.15	7.08	Inhibition of USP36 (unknown origin) using Ub-Rho measured after 0.5 to 3 hrs	Homo sapiens	Scientific Literature
CHEMBL5 172138	276.25	2.51	19.58	0.35	2.9	7.4	Inhibition of USP36 (unknown origin) using Ub-Rho measured after 0.5 to 3 hrs	Homo sapiens	Scientific Literature
CHEMBL5 185273	426.83	3.99	11.89	0.23	1.09	6.99	Inhibition of USP36 (unknown origin) using Ub-Rho measured after 0.5 to 3 hrs	Homo sapiens	Scientific Literature
CHEMBL5 426200	462.46	4.06	13.82	0.27	2.33	6.38	Inhibition of human USP36 (81 to 461 residues) using Ub-Rh110MP as substrate by fluorescence based assay	Homo sapiens	SureChEMBL Patent Bioactivity Data
CHEMBL5 431379	459.32	4.54	13.24	0.28	1.54	5.98	Inhibition of human USP36 (81 to 461 residues) using Ub-Rh110MP as substrate by fluorescence based assay	Homo sapiens	SureChEMBL Patent Bioactivity Data
CHEMBL5 303447	462.46	4.06	14.49	0.29	2.64	6.7	Inhibition of human USP36 (81 to 461 residues) using Ub-Rh110MP as substrate by fluorescence based assay	Homo sapiens	SureChEMBL Patent Bioactivity Data

(iii) Briefly discuss (100 words max) the **druggability** of your novel protein based on:
- Presence of well-defined pockets (output of tools like CASTpFold), - Existence of known inhibitors for related proteins (your search of ChEMBEL), - Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

The novel protein shows good druggability. CASTpFold analysis shows a well-defined pocket (area 497.9 A^2, volume 1078.5 A^3), indicating possible space for binding. A model constructed with AlphaFold and visualized in PyMOL shows much of the sequence (b>95%) is conserved, supporting conservation of binding sites across homologs. ChEMBL shows assays demonstrating inhibition of a similar protein in humans (USP36). USP36 stabilizes c-Myc through deubiquitination, promoting ribogenesis in cancers (Sun 2015). Targeting the novel protein may have a similar effect in plants, or up-regulation might be of interest in the case that plant growth is encouraged.

Works cited:

1. Sun, Xiao-Xin et al. "The nucleolar ubiquitin-specific protease USP36 deubiquitinates and stabilizes c-Myc." *Proceedings of the National Academy of Sciences of the United States of America* vol. 112,12 (2015): 3734-9. doi:10.1073/pnas.1411713112

2. S. Reinbothe, C. Reinbothe, J. Lehmann, W. Becker, K. Apel, & B. Parthier, JIP60, a methyl jasmonate-induced ribosome-inactivating protein involved in plant stress reactions., Proc. Natl. Acad. Sci. U.S.A. 91 (15) 7012-7016, https://doi.org/10.1073/pnas.91.15.7012 (1994).

Scoring Rubric: [60 total points available]

Q1 (4 points)

Protein name 1 Species 1 Accession number 1

Function known 1

Q2 (6 points)

Blast method 1 Database searched 1 Limits applied 1

Search output list (top hits) 1 Alignment of choice 1

Evalue and other alignment stats 1

Q3 (3 points)

Protein sequence of choice matches Subject above 1

Name in header 1 Species 1

Q4 (3 point)

Blastp output list with identities & Evalue 1 Top alignment shown with alignment statistics 1 Results indicates a "novel" gene found 1

Q5 (3 points)

MSA labeled with useful names 1 MSA trimmed appropriately (i.e. no gap overhangs) 1 Pasted MSA fits report page width (i.e. font, format) 1

Q6 (1 point)

Figure illustrates sequence clustering pattern 1

Q7 (10 points)

Heatmap figure included in report 5 Heatmap is legible (i.e. no labels obscured) 5

Q8 (10 points)

PDB identifiers from multiple species reported 5 Annotation of PDB source, resolution and technique 4 Annotation of Evalue and Sequence Identity 1

Q9 (10 points)

Structure figure provided 2 Uses white background for molecular figure 1 Figure of high resolution (i.e. not just snapshot) 1 Conserved residues as spacefill 3 Protein cartoon colored by pLDDT quality score 3

Q10 (10 points)

i) Binding site image, volume and area. 3 ii) Evidence of ChEMBEL searches 1 iii) Druggability discussion 6